

Design Thinking, Innovation and Entrepreneurship (25HU102)
Question Bank
Unit-2 Design Thinking Framework

Q1. What are the five phases of the Design Thinking process?

A1. Empathise, Define, Ideate, Prototype, and Test.

Q2. Which phase focuses on understanding the needs and challenges of users?

A2. The Empathise phase.

Q3. What is created in the Prototype phase?

A3. Tangible models, mockups, or low-fidelity versions of potential solutions.

Q4. Why is the Design Thinking process described as iterative?

A4. Because teams may revisit earlier stages as new insights emerge to refine and improve solutions.

Q5. Which key characteristic ensures ideas are explored through hands-on experimentation?

A5. Action-Oriented nature of Design Thinking.

Q6. What actions are taken in the Define phase?

A6. Insights from the Empathise phase are synthesized to form a clear problem statement, reframed from a user-centric perspective.

Q7. How is collaboration encouraged in the Design Thinking framework?

A7. By involving teams from diverse backgrounds, fostering knowledge exchange, and combining multiple perspectives for innovative solutions.

Q8. What is the main focus of the Ideate phase?

A8. To generate a wide range of creative and innovative solutions in a judgment-free brainstorming environment.

Q9. A company designs a new app, but users find it confusing. At which stage of Design Thinking should they return, and why?

A9. They should return to the Empathise and Define phases to better understand user needs and redefine the problem, ensuring the solution truly addresses user pain points.

Q10. Why is user feedback critical in the Test phase?

A10. Because it helps identify strengths, weaknesses, and areas for improvement, allowing the solution to be refined and made more user-friendly and effective.

Q11. If a team jumps directly to prototyping without proper user research, what risk do they face?

A11. They risk creating solutions that are not aligned with real user needs, leading to ineffective or irrelevant outcomes.

Q12. Define “user-centric” in the context of Design Thinking.

A12. User-centric means keeping the entire process focused on understanding and solving the actual needs, challenges, and motivations of the end-users.

Q13. How does iteration improve the final solution?

A13. Iteration allows continuous refinement through feedback, ensuring the solution evolves to become more innovative, effective, and user-friendly.

Q14. What is the main focus of the Empathise phase in Design Thinking?

A14. To understand the needs, motivations, and challenges of users.

Q15. Mention two actions taken during the Empathise phase.

A15. Conducting user research and interviews.

Q16. Which research method helps in gaining direct insights into user experiences?

A16. Observations.

Q17. Why is the Empathise phase considered important in the Design Thinking process?

A17. Because it helps designers gain deep insights into users’ perspectives, ensuring solutions are user-centred.

Q18. A team designing a healthcare app spends time shadowing doctors and patients. Which Design Thinking phase are they working in?

A18. Empathise phase.

Q19. What is the main focus of the Define phase in Design Thinking?

A1. To synthesize insights from the Empathise phase into a clear and concise problem statement.

Q20. What does the Define phase help prevent before moving toward solutions?

A2. It prevents rushing to solutions without a thorough understanding of the problem.

Q21. From which point of view should the challenge be reframed during the Define phase?

A3. From a user-centric point of view.

Q22. Why is creating a problem statement important in the Define phase?

A4. Because it ensures that the team has a well-understood and focused challenge to address, guiding the ideation process effectively.

Q23. A team realizes they need to reframe their challenge to focus on what users truly need. Which Design Thinking phase are they working in?

A5. The Define phase.

Q24. What is the main focus of the Ideate phase in Design Thinking?

A1. To generate a wide range of creative solutions to the defined problem.

Q25. Name one key activity carried out in the Ideate phase.

A2. Brainstorming.

Q26. Why is a judgment-free environment important in the Ideate phase?

A3. It encourages bold and innovative ideas to surface without fear of criticism.

Q27. What type of thinking is promoted during the Ideate phase?

A4. Divergent and creative thinking to explore multiple possible solutions.

Q28. A design team arranges a brainstorming session where no idea is dismissed immediately. Which Design Thinking phase are they practising?

A5. The Ideate phase.

Q29. What is the main focus of the Prototype phase in Design Thinking?

A29. To turn abstract ideas into tangible models or low-fidelity versions of potential solutions.

Q30. Mention two examples of prototypes created during this phase.

A30. Mockups and sketches.

Q31. Why are prototypes important in the Design Thinking process?

A31. They help visualize and explore concepts, making ideas more concrete and testable.

Q32. At which stage of Design Thinking are basic versions of solutions created to bring ideas to life?

A32. The Prototype phase.

Q33. A team develops a simple cardboard model of a product idea. Which Design Thinking phase are they working in?

A33. The Prototype phase.

Q34. What is the main focus of the Test phase in Design Thinking?

A1. To evaluate the prototypes with actual users to gather feedback and identify areas for improvement.

Q35. How are prototypes tested during this phase?

A2. By testing them in real-world situations.

Q36. What should be done with the feedback collected in the Test phase?

A3. It should be used to refine the solution and, if necessary, return to earlier phases for iteration.

Q37. Why is user feedback critical in the Test phase?

A4. Because it reveals what works, what doesn't, and guides improvements to make the solution more effective and user-friendly.

Q38. A design team collects feedback from users after trying out their app prototype and decides to make changes. Which Design Thinking phase are they in?

A5. The Test phase.

Q19. Apply the 5-step process of Design Thinking to the following case study.

- 1. Lucky Iron Fish**
- 2. Embrace Infant Warmer**

Q&A: Embrace Infant Warmer Case Study

Q1. How many premature and underweight babies are born globally each year?

A1. About 15 million.

Q2. What is the main cause of death among premature babies in developing countries like India?

A2. Hypothermia (inability to maintain body temperature).

Q3. How much does a standard incubator cost, and why is it unsuitable in rural India?

A3. Around \$20,000; it is too expensive, requires electricity, and is not feasible in rural areas.

Q4. What was the final product developed by the innovators?

A4. The Embrace Infant Warmer, resembling a baby sleeping bag.

Q5. Approximately how much does the Embrace Infant Warmer cost?

A5. About \$25.

B. Process-Oriented Questions (by Design Thinking stages)

Q6. What research methods did student innovators use in the Empathize phase?

A6. They visited rural India, interacted with mothers, doctors, and health workers, and studied the challenges faced in newborn care.

Q7. What challenge statement was defined in the Define phase?

A7. *“How might we design a baby-warming solution that is affordable, portable, electricity-independent, and easy to use in rural India?”*

Q8. Mention one idea explored during the Ideation phase before the final concept was chosen.

A8. Hot-water bottles, heating lamps, or blankets.

Q9. What material innovation was included in the Prototype phase?

A9. A phase-change material pouch that maintained 37°C for 4–6 hours.

Q10. What was the feedback during the Test phase?

A10. It was easy to use, kept babies’ temperatures stable, and was adaptable in clinics and homes.

C. Application & Critical Thinking Questions

Q11. Why can the Embrace Infant Warmer be considered a frugal innovation?

A11. Because it provides a life-saving solution at just \$25 compared to \$20,000 incubators, making it affordable, simple, and accessible.

Q12. How did empathy play a role in the success of this innovation?

A12. By directly engaging with mothers and health workers in rural India, the team understood real challenges and designed a culturally acceptable, user-friendly solution.

Q13. Compare the Embrace Infant Warmer with incubators in terms of cost and usability.

A13. Incubators cost ~\$20,000, require electricity, and are suited for advanced hospitals, whereas the Embrace warmer costs ~\$25, works without electricity, and can be used in rural homes and clinics.

Q14. In which way did field trials influence the design iteration?

A14. Feedback from mothers and health workers helped improve the durability and cultural fit of the warmer.

D. Outcome & Impact Questions

Q15. In how many countries is the Embrace Infant Warmer used, and how many infants has it saved?

A15. Used in 20+ countries, saving over 300,000 infants.

Q16. Name two global recognitions received by the Embrace Infant Warmer.

A16. Awards from the World Economic Forum and the Skoll Foundation (also recognition at TED).

Q17. What broader lesson does the Embrace Infant Warmer case study teach about design thinking?

A17. That empathy-driven, user-centered design can create impactful, affordable, and scalable solutions to global healthcare challenges.

Q&A: Lucky Iron Fish Case Study

Q1. What health problem was the Lucky Iron Fish designed to address?

A1. Severe iron deficiency and anaemia, especially among women and children in rural Cambodia.

Q2. Why were iron pills and supplements not a sustainable solution?

A2. They were expensive, had side effects, and were not consistently used.

Q3. In the Prototype phase, why did villagers reject the first iron ingots?

A3. Because they looked like scrap metal and were not culturally meaningful.

Q4. What shape was chosen for the redesigned prototype, and why?

A4. A fish, as it symbolized luck and prosperity in Cambodian culture.

Q5. For how many years can one Lucky Iron Fish be reused?

A5. For more than 5 years.

B. Process-Oriented Questions (Design Thinking Stages)

Q6. What insight did the team gain during the Empathize phase?

A6. Locals suffered from fatigue and poor health due to anaemia and had a distrust of Western medicines, leading to low adoption of iron pills.

Q7. State the challenge framed in the Define phase.

A7. *“How might we deliver iron in a simple, affordable, and culturally acceptable way to fight anaemia?”*

Q8. What three ideas were explored during the Ideate phase, and which one was selected?

A8. Food fortification (too costly), iron cookware (not practical), and iron blocks for cooking pots (selected).

Q9. How was cultural empathy applied in the Prototype phase?

A9. By redesigning the ingot into the culturally significant shape of a fish, increasing acceptance.

Q10. What evidence proved the success of the Test phase?

A10. Clinical trials showed significant improvements in iron levels and reduction in anaemia among users.

C. Application & Critical Thinking Questions

Q11. Why can the Lucky Iron Fish be considered a sustainable solution?

A11. It is affordable, reusable for years, easy to integrate into daily cooking, and has no harmful side effects.

Q12. How did the Lucky Iron Fish project demonstrate the importance of iterative prototyping?

A12. The initial design was rejected, but redesigning into a culturally meaningful shape (fish) made it widely accepted, showing how iteration improved success.

Q13. In what way did cultural factors influence the success of the Lucky Iron Fish?

A13. The fish design connected with local beliefs of luck and prosperity, making people more willing to adopt and use it.

D. Outcome & Impact Questions

Q14. Beyond Cambodia, where has the Lucky Iron Fish made an impact?

A14. It has been adopted in multiple countries globally, benefiting millions of people.

Q15. How did the Lucky Iron Fish evolve into a social enterprise?

A15. By selling the product globally and using profits to fund free distribution in needy communities.

Q16. What broader lesson does the Lucky Iron Fish case teach about design thinking?

A16. That cultural empathy, combined with iterative prototyping and user feedback, is key to creating impactful and scalable social innovations.
